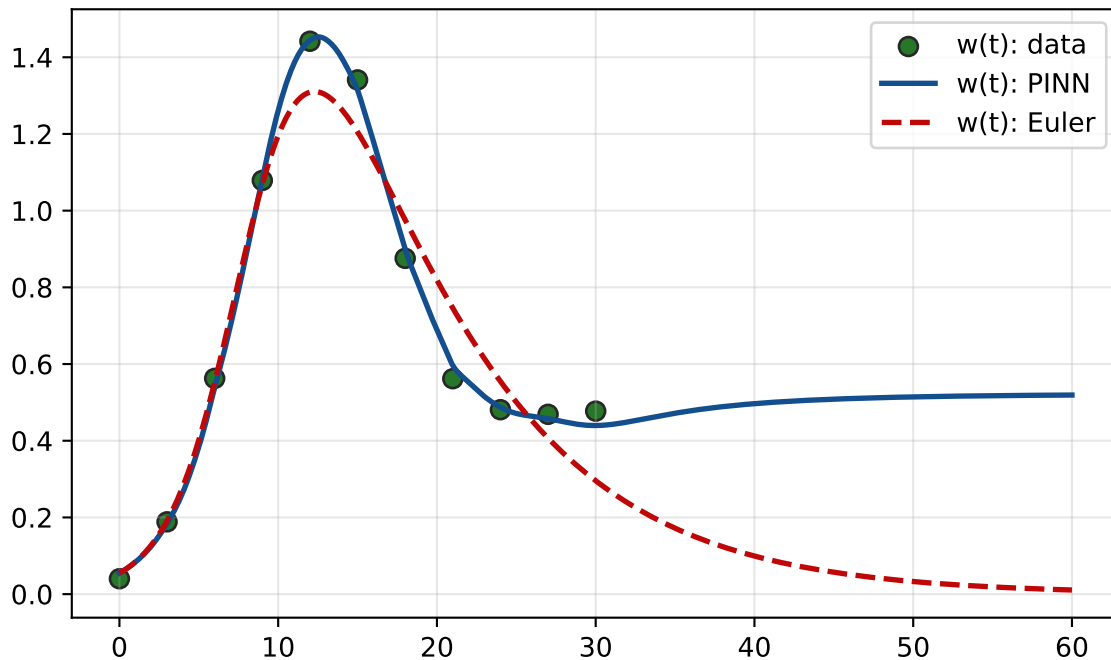
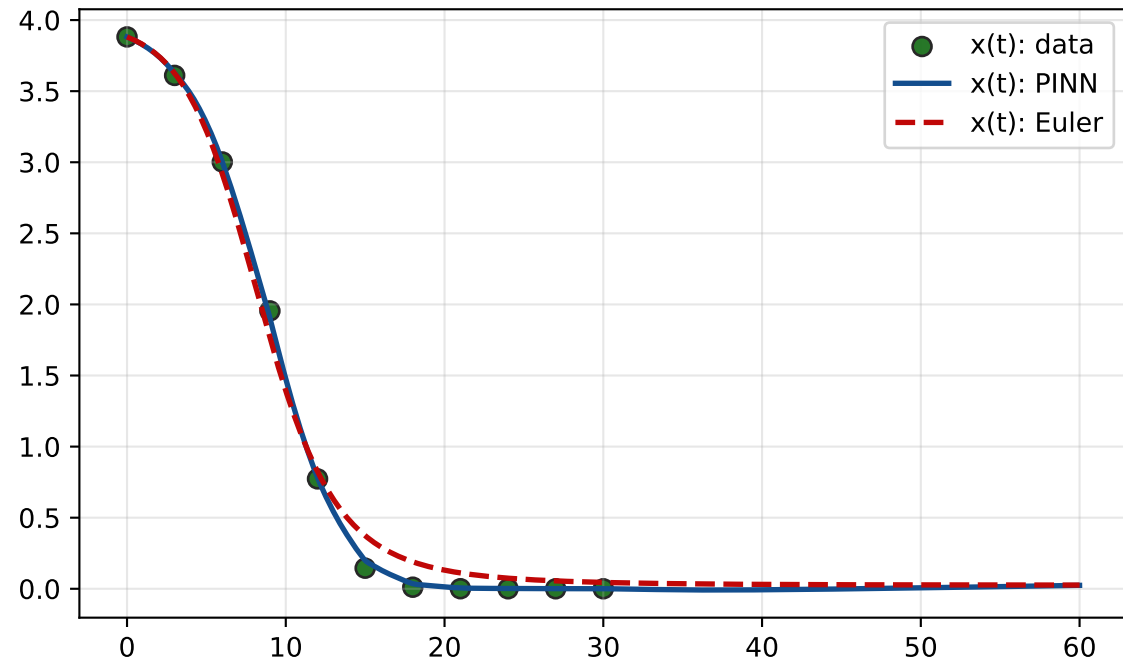


# E2 | PINN vs Euler

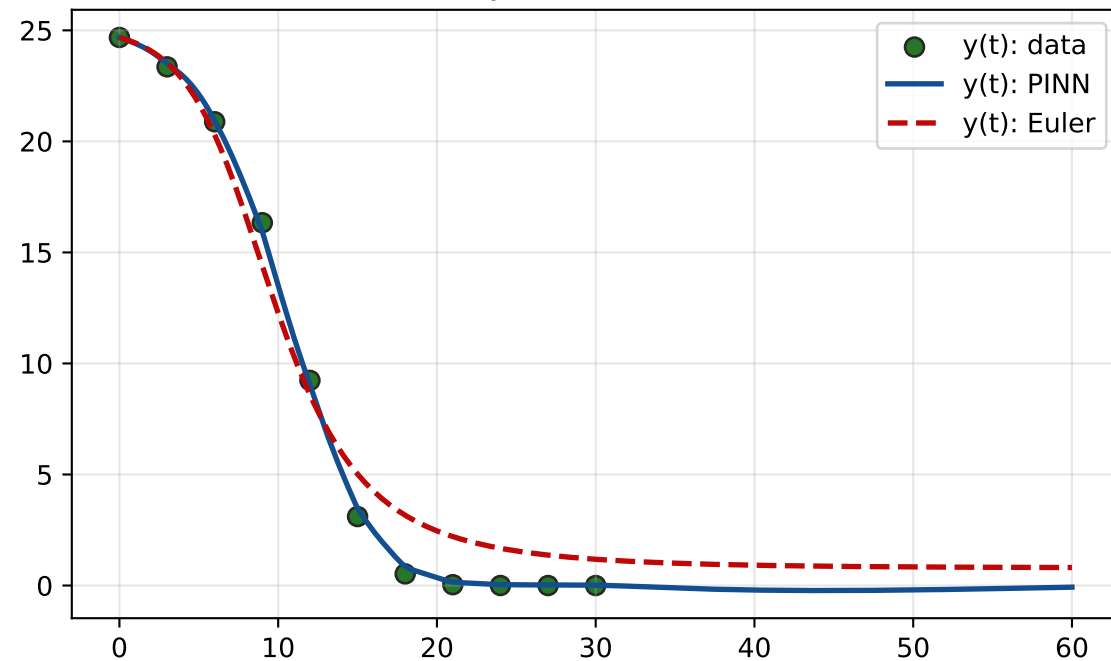
w(t): Biomass



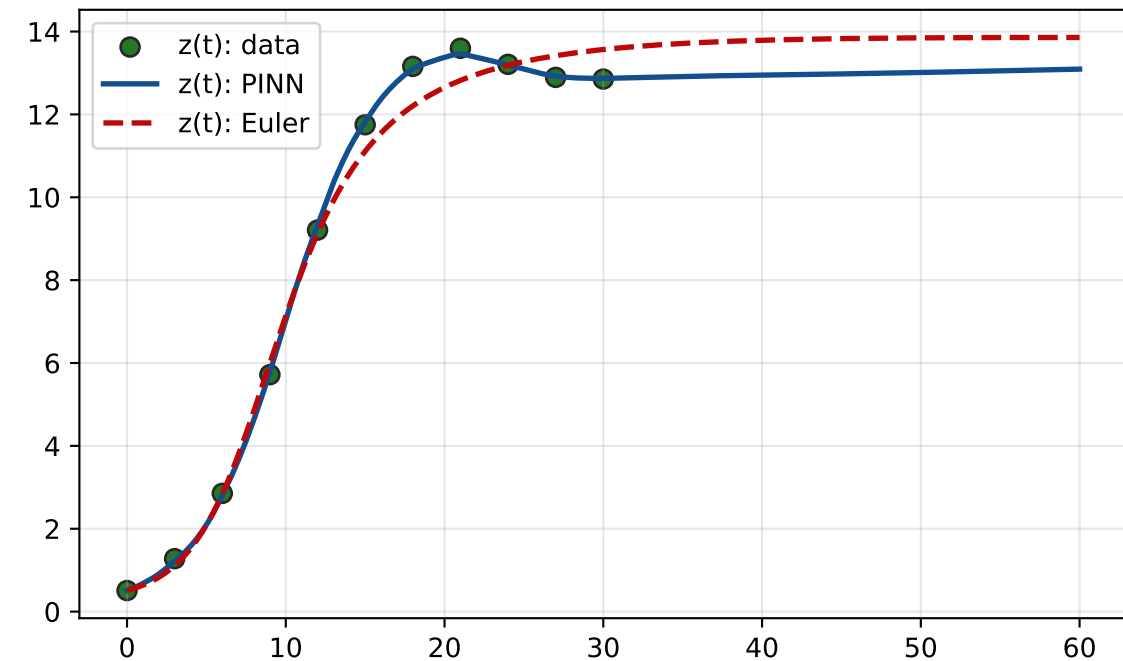
x(t): Glucose



y(t): Fructose



z(t): Ethanol



E2

R2=0.9860, R2\_adj=0.9832, RSS=2.905e+01, AIC=-1.151e+00

cond(JTJ)=1.15e+05, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 1.553366e-02 | 2.501707e-03 | 5.068939e-03 | 1.046472e-02  | 2.060260e-02 | 3.271827e-07 |
| p2    | 7.058849e-02 | 1.149990e-01 | 2.330102e-01 | -1.624217e-01 | 3.035986e-01 | 5.430912e-01 |
| p3    | 7.650440e-03 | 5.670296e-03 | 1.148911e-02 | -3.838672e-03 | 1.913955e-02 | 1.854675e-01 |
| p4    | 2.067790e-01 | 9.899244e-02 | 2.005777e-01 | 6.201294e-03  | 4.073568e-01 | 4.365845e-02 |
| p6    | 1.401536e-01 | 4.301427e-02 | 8.715519e-02 | 5.299841e-02  | 2.273088e-01 | 2.405532e-03 |
| p8    | 6.321669e-02 | 2.011096e-02 | 4.074867e-02 | 2.246803e-02  | 1.039654e-01 | 3.284204e-03 |
| p9    | 1.007740e-04 | 6.185466e-03 | 1.253294e-02 | -1.243217e-02 | 1.263372e-02 | 9.870889e-01 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 5.467984e-02 |
| x0 (fixed)     | 3.881512e+00 |
| y0 (fixed)     | 2.468014e+01 |
| z0 (fixed)     | 5.083666e-01 |
| W0 (fixed)     | 0.000000e+00 |